

Crosslinked Polymers

Thermosets

Thermosets

Thermosets: A Different State of Polymer Matter

Once curing is complete, the material becomes one macroscopic network.

Heating may soften the material somewhat below degradation, but it cannot be melted (infusible) and reprocessed.

In comparison with thermoplastics, this explains why thermosets dominate applications requiring dimensional stability, high-temperature performance, chemical resistance, and structural integrity.

Thermosets

From Monomers to Networks

Three events (may) occur during synthesis:

- conversion rises as reactive groups react;
- gelation occurs when a sample-spanning network appears and the material loses its ability to flow as a liquid;
- vitrification may occur if the glass transition temperature rises above the curing temperature.

Thermosets

Cure Kinetics and Network Development

Connectivity ↑

→ Molecular weight ↑

→ Mobility ↓

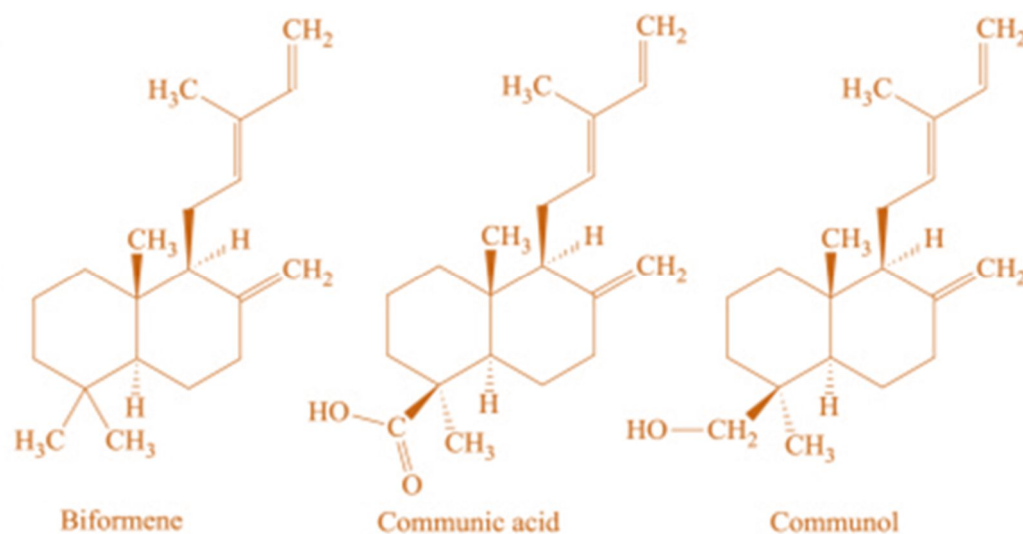
Thermosets

A Short Historical Perspective

- *Ever since plants turn into sediments* **Amber**



formed by free radical polymerization of terpenoid precursors



<https://balticproud.com/blog/what-color-is-amber-timeless-gemstone-of-the-baltic-sea/>

Thermosets

Phenolics such as Bakelite were the first industrial thermosets.

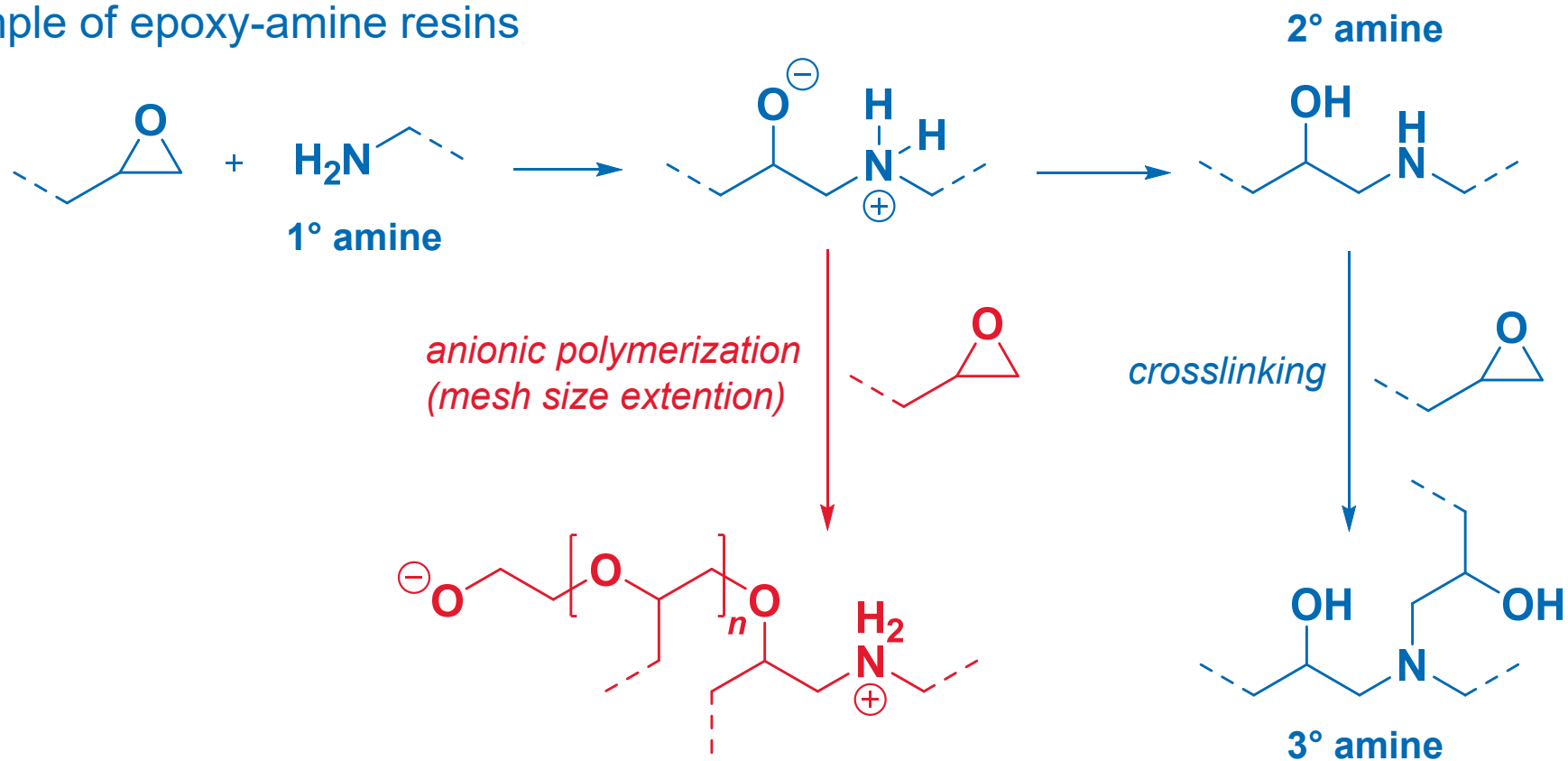
In the following the subsequently most relevant thermosets are reviewed:

- **Epoxies** — benchmark engineering thermosets
- **Polyurethanes** — polymer materials with architecture-dependent versatility
- **Unsaturated Polyesters** — commodity composite matrices
- **Vinyl Esters** — upgraded polyester systems
- **Bismaleimides** — aerospace thermosets
- **Cyanate Esters** — thermosets for electronics and low dielectric applications
- **Polyimides** — thermosets with extreme thermal performance
- **Benzoxazines** — next-generation thermosets

Thermosets

Epoxy Resins: The Benchmark Thermoset

Example of epoxy-amine resins

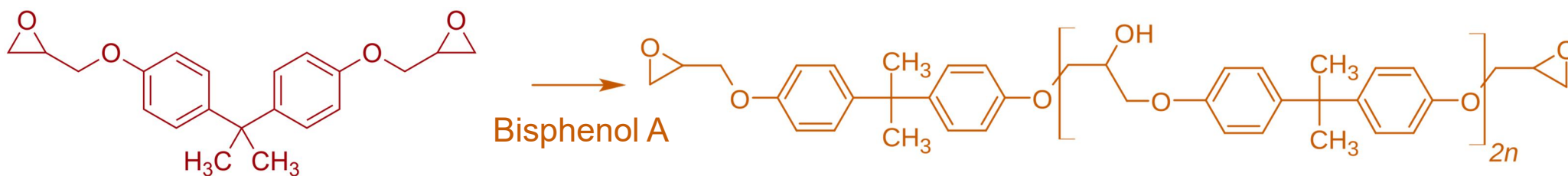


“autocatalytic reaction”

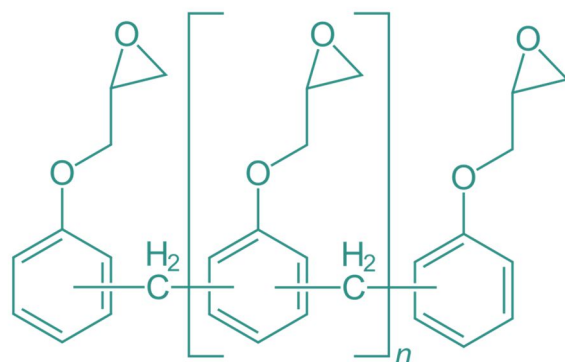
Thermosets

Epoxy Resins: The Benchmark Thermoset

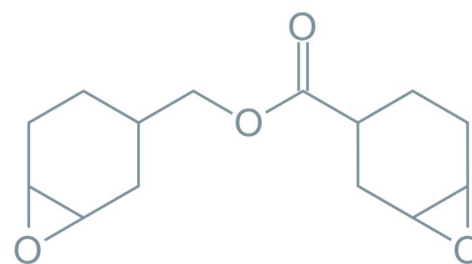
Epoxide + Hardener



DGEBA



Epoxyphenol-Novolak



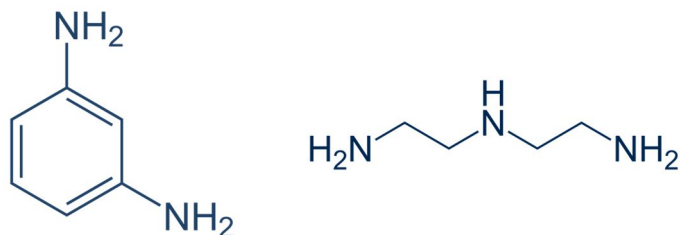
Cycloaliphatic epoxides

Thermosets

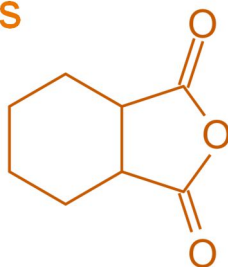
Epoxy Resins: The Benchmark Thermoset

Epoxide + Hardener

- amines



- anhydrides



- phenols, thiols, isocyanates

By varying functionality, hardener type, and stoichiometry, properties can be tuned, e.g., modulus, toughness, glass transition temperature, cure schedule.

Thermosets

Polyurethanes: A Chemistry, Not a Material Class

One chemistry → many materials

Polyurethanes are **polymeric** materials with **urethane** linkages.

But polyurethanes are *not* one type of materials.

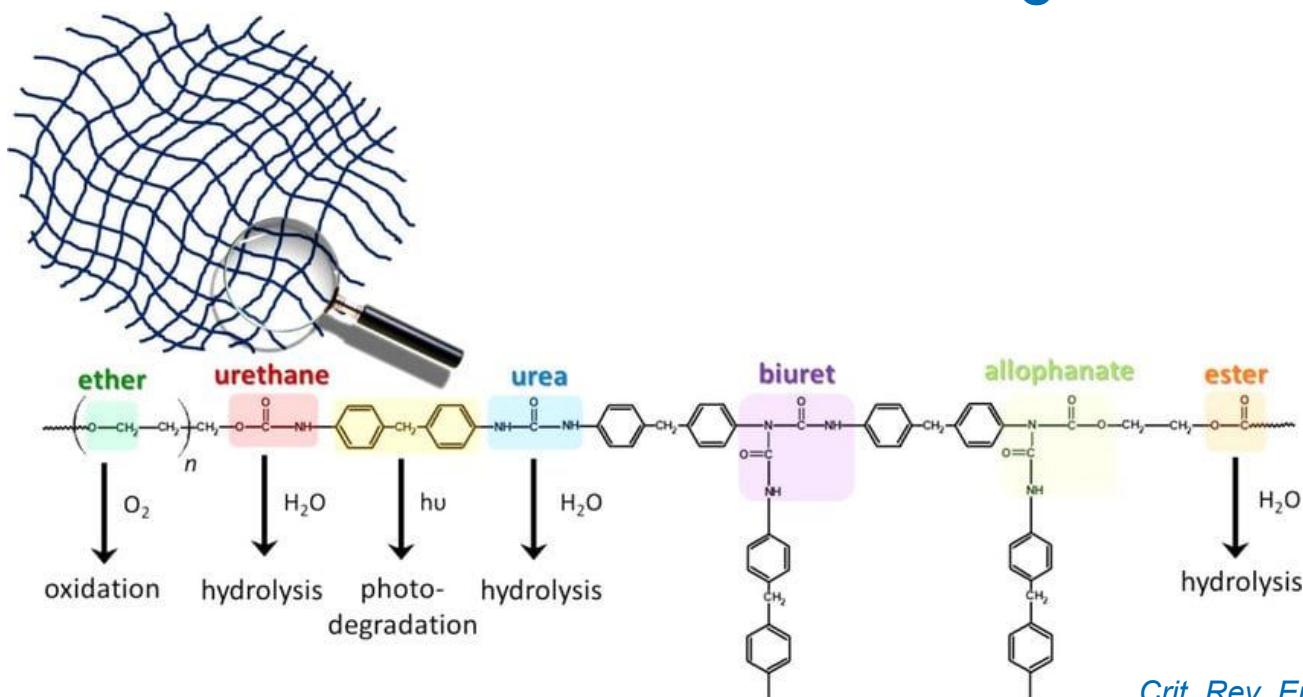
They can be any one of the following material type:

- thermoplastics
- elastomers
- foams
- thermosets

Thermosets

Polyurethanes Thermosets

Isocyanate + Polyol



Crit. Rev. Environ. Sci. Technol. **2022**, 52, 157

Thermosets

Polyurethanes Thermosets

Polyurethane Chemistry: More Than Just Urethanes

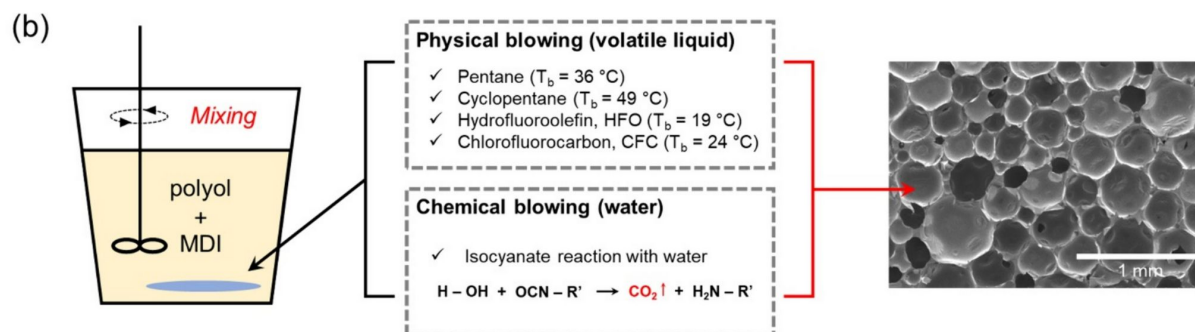
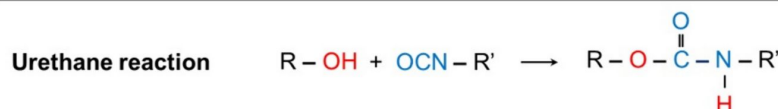
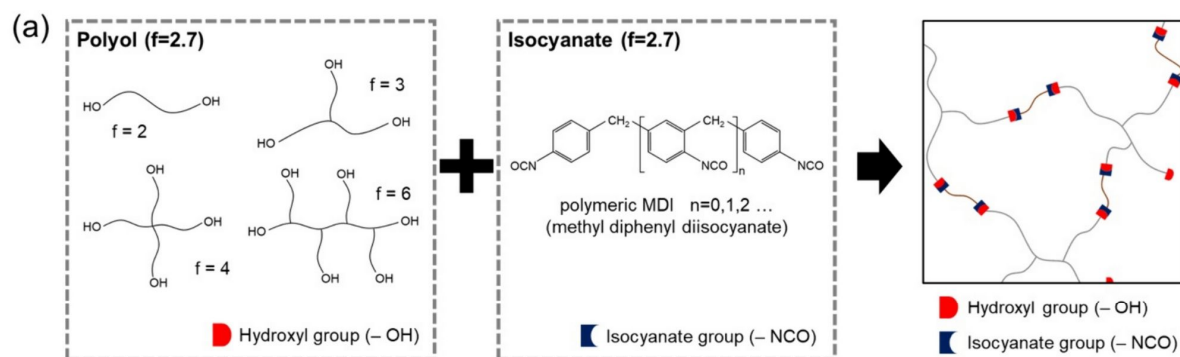


Polyurethane chemistry is a network of reactions, not a single reaction.

Thermosets

Polyurethane Foams

These are typically crosslinked network, with a cellular structure and a low density.

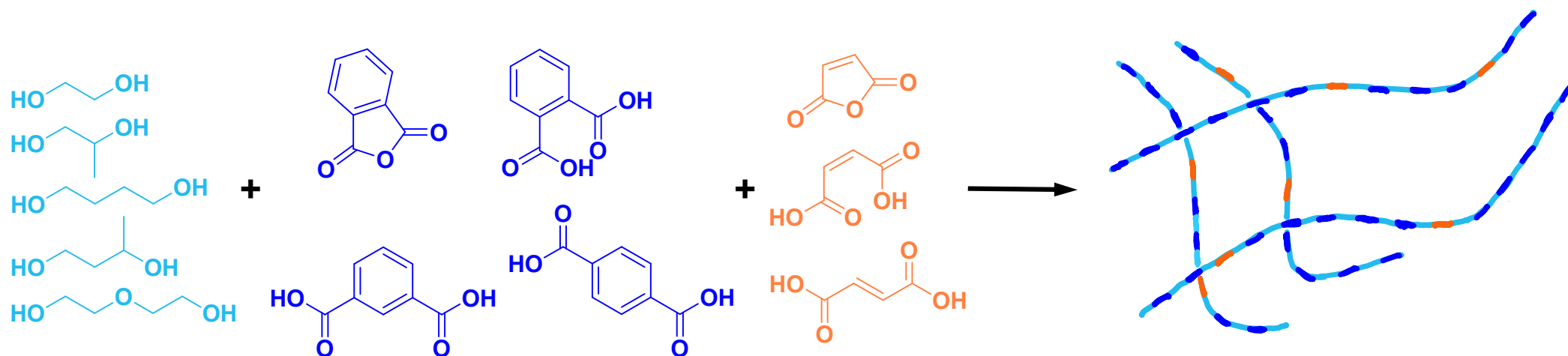


Crit. Korean J. Chem. Eng. **2024**, *41*, 3139

Thermosets

Unsaturated Polyester Resins

The Workhorse of Fiber-Reinforced Plastics

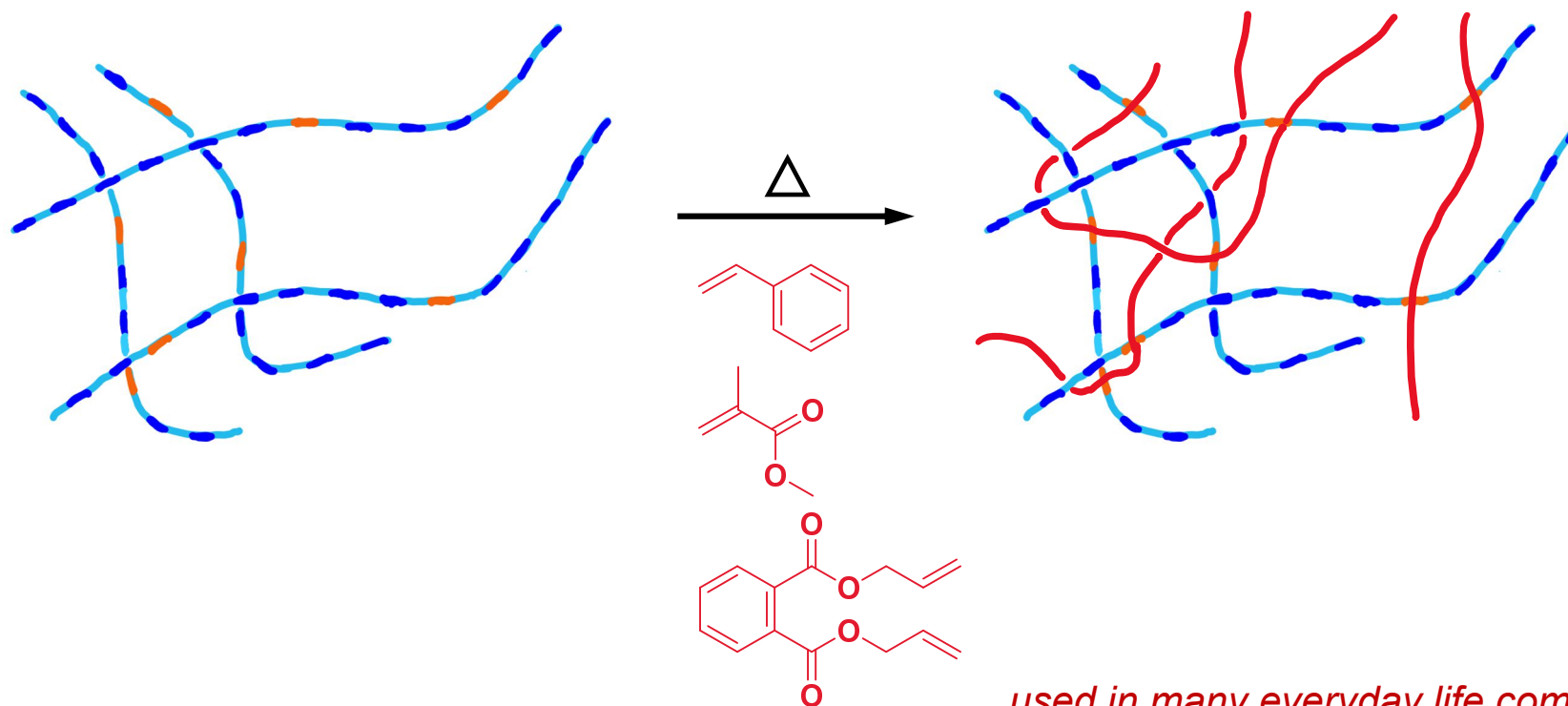


- better toughness
- better chemical resistance
- lower shrinkage

Thermosets

Unsaturated Polyester Resins

Radical Crosslinking in Polyester Resins



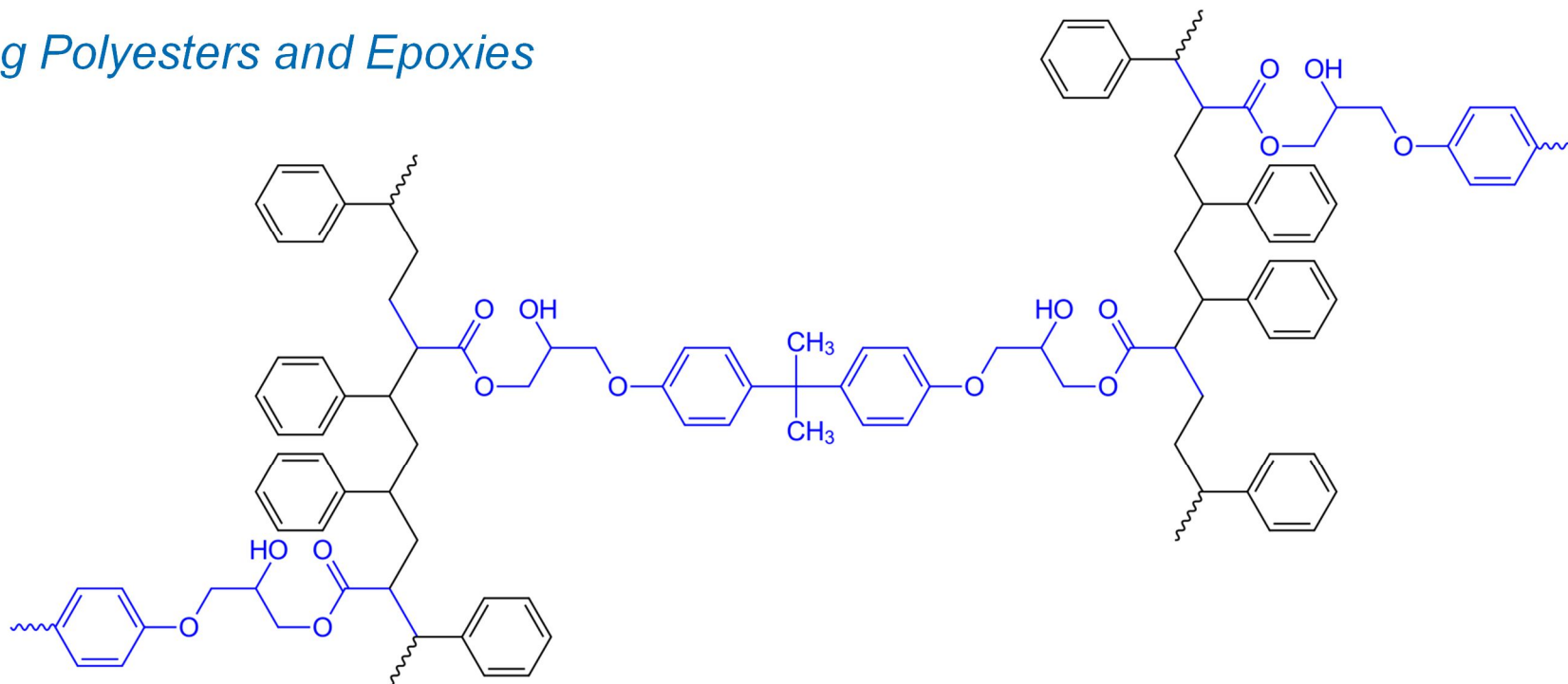
used in many everyday life composite materials, e.g., automotive body components, boat hulls, tanks, pipes, construction panels

Thermosets

Vinyl Ester Resins

Bridging Polyesters and Epoxies

<https://de.wikipedia.org/wiki/Vinylesterharz>



vs unsaturated polyesters

- better toughness
- better chemical resistance
- lower shrinkage

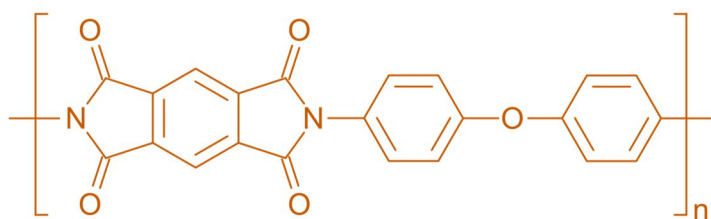
vs epoxies

- lower cost
- simpler processing

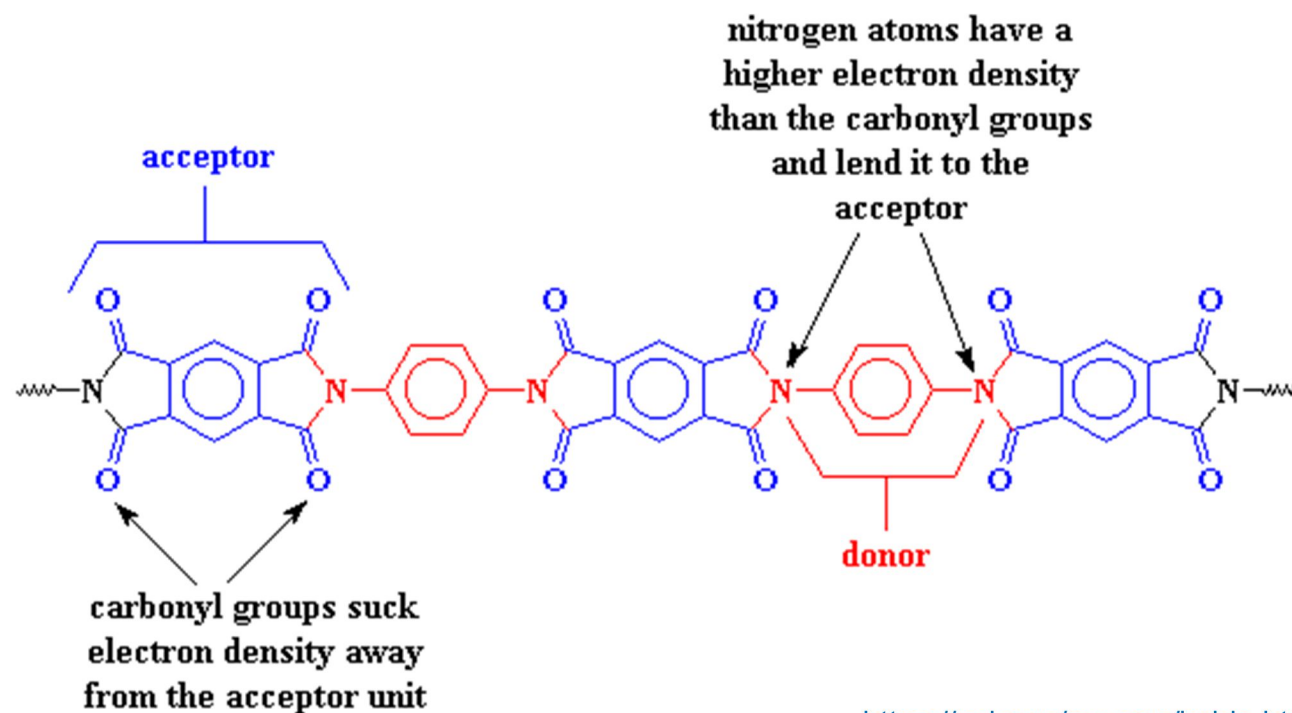
Thermosets

Bismaleimide (BMI) Resins

vs regular polyimides



Kapton® (DuPont)



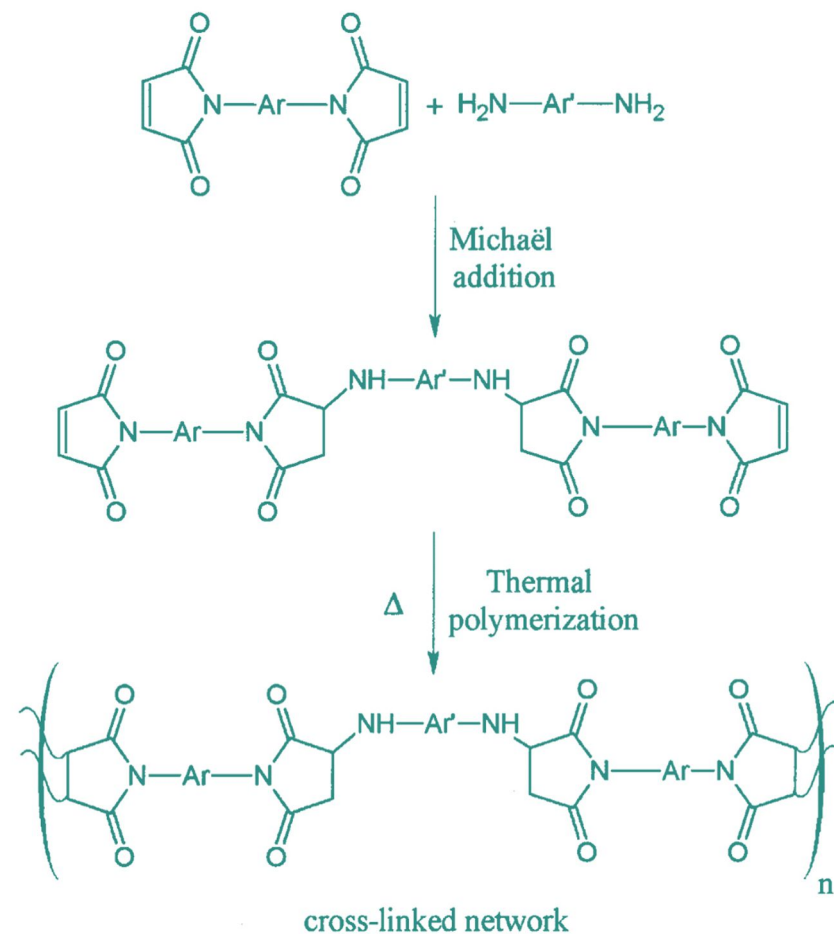
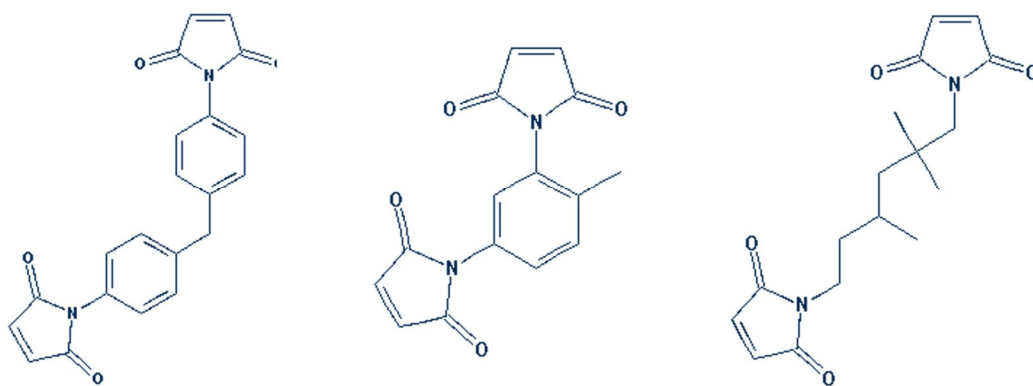
<https://en.wikipedia.org/wiki/Kapton>

<https://pslc.ws/macrog/imide.htm>

Thermosets

Bismaleimide (BMI) Resins

High-Temperature Thermosets
for aerospace and electronics

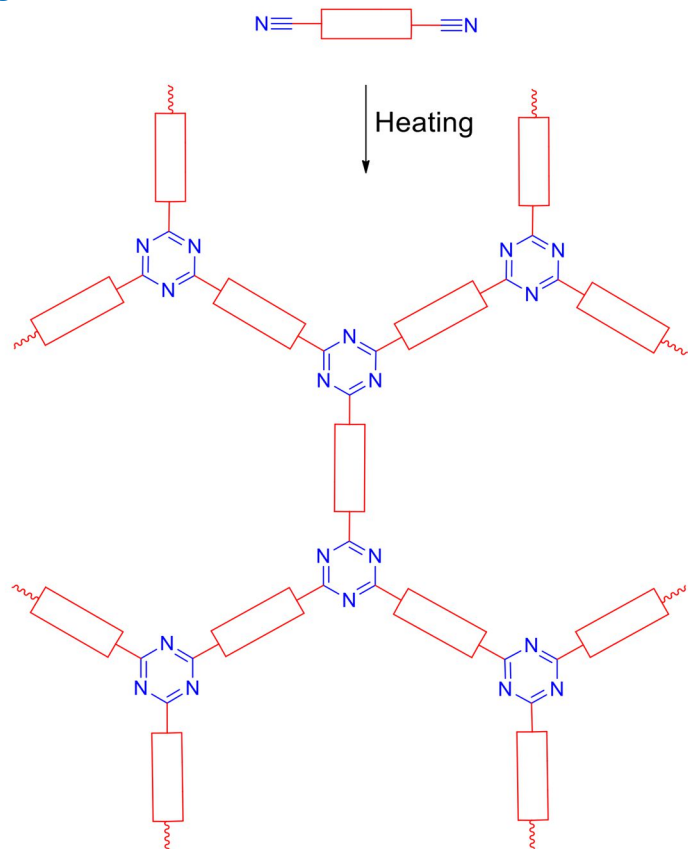


<https://polymerinnovationblog.com/beyond-epoxy-high-performance-thermosets-part-two-bismaleimide-cure-chemistry/>

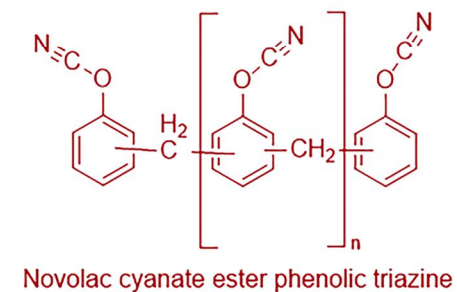
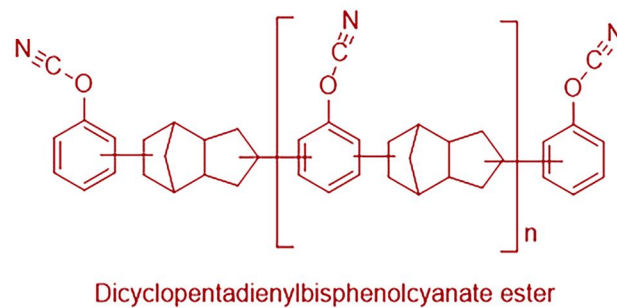
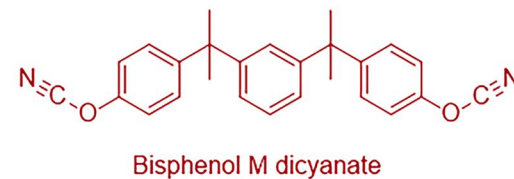
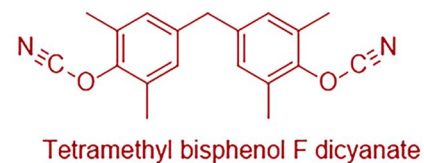
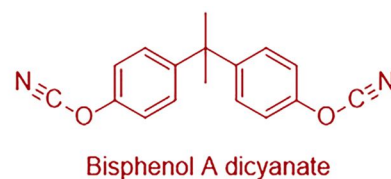
Thermosets

Cyanate Ester Resins

Cyanate → Triazine



commercial CE "monomers"



Thermosets

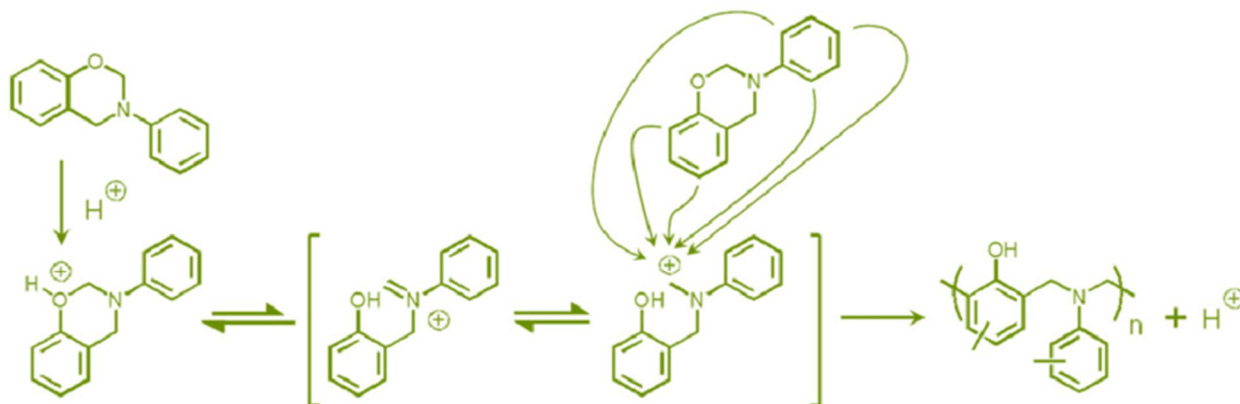
Benzoxazines

The Emerging Thermoset Family

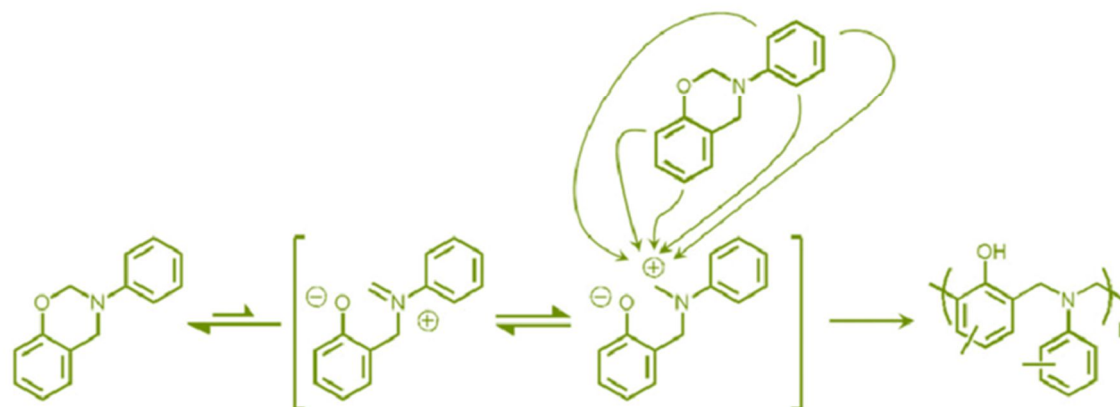
Ring-Opening Mechanism

Possibly no catalyst required

a) Thermally accelerated (or activated) polymerization mechanism



b) Intrinsic thermal ring-opening polymerization mechanism



Polym. Degrad. **2020**, 179, 109279

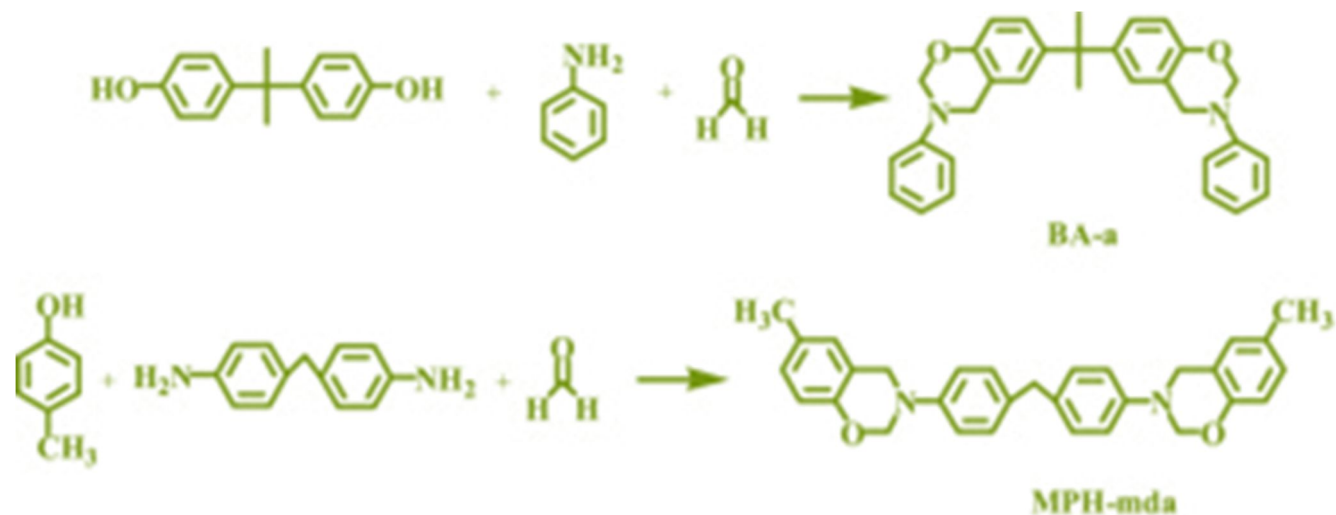
Thermosets

Benzoxazines

The Emerging Thermoset Family

Ring-Opening Network Formation

- Low shrinkage
- High T_g

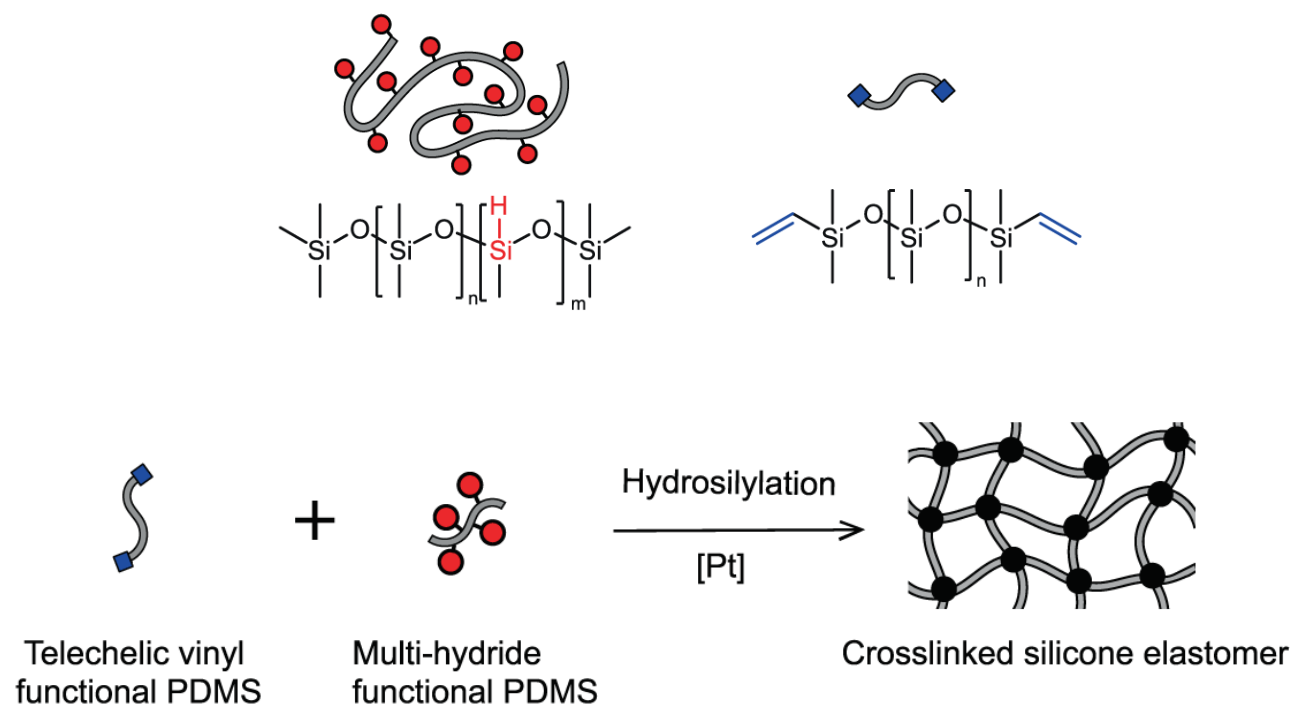


Polym. Int. **2022**, 71, 583

Thermosets

[Lightly Crosslinked Thermosets] → Elastomers

Silicone Elastomers (PDMS)



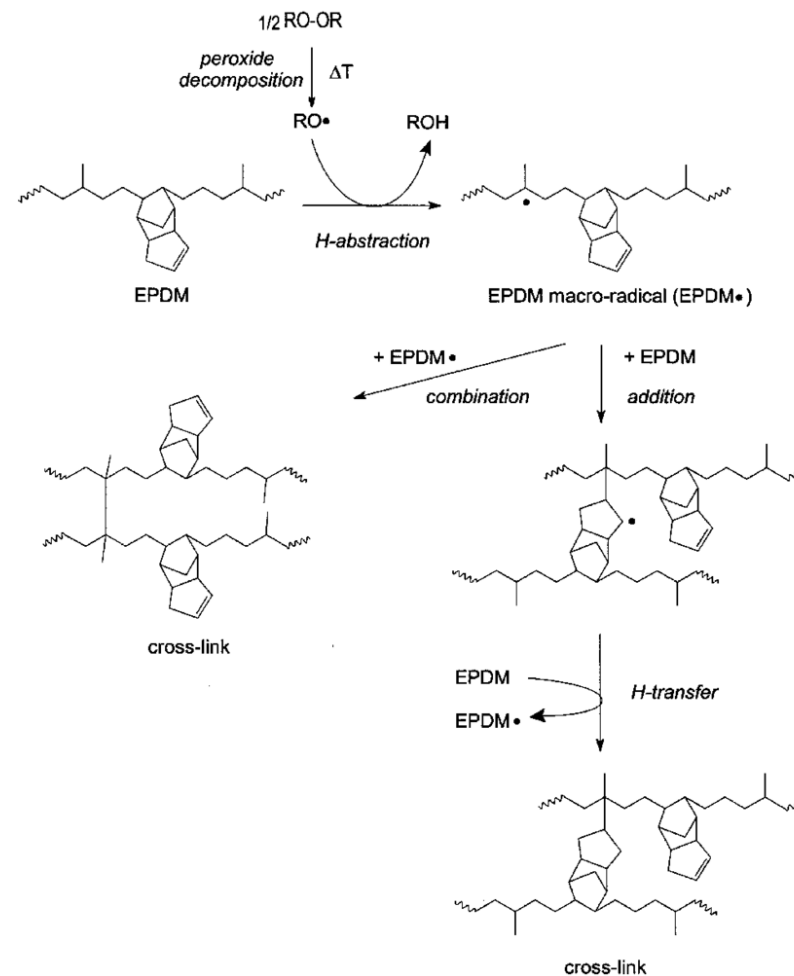
Thermosets

[Lightly Crosslinked Thermosets] → Elastomers

Ethylene Propylene Diene (EPDM) Rubber

Peroxide curing

(possibility to include sulfur in the process)



Chemistry of EPDM Cross-linking

in *KGK Kautschuk Gummi Kunststoffe* 55. Jahrgang, Nr. 4/2002